

Niloofer Miresghallah

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RESEARCH INTERESTS

- Privacy, contextual integrity & information-flow norms
- Memorization, membership inference & data integrity
- AI for science: chemistry & drug discovery
- AI, mental health & user wellbeing

POSITIONS

Assistant Professor

Carnegie Mellon University

Aug 2025–Present

Joint appointment: Language Technologies Institute (LTI) & Engineering and Public Policy (EPP); courtesy appointment in Software and Societal Systems (S3D); core member of CyLab.

Founding Member of Technical Staff

humans& ai

Nov 2025–Nov 2026

User Model Development and Evaluation, RL Environment Development

Research Scientist

Meta AI (FAIR Alignment: Privacy & Security; collab. with FAIR Chemistry)

May 2025–Nov 2025

Privacy-preserving AI systems and LLM safety.

Postdoctoral Scholar

University of Washington

May 2023–May 2025

Advisors: Yejin Choi, Yulia Tsvetkov.

Research Intern / Part-time Researcher

Microsoft (Semantic Machines; MSR Algorithms; MSR Language & Intelligent Assistance) *Jun 2020–Jul 2023*

Mentors: Richard Shin, Yu Su, Tatsunori Hashimoto, Jason Eisner, Sergey Yekhanin, Arturs Backurs, Dimitrios Dimitriadis, Robert Sim, Huseyin Inan.

Research Intern

Western Digital Co. Research and Development

Summer 2019

EDUCATION

Ph.D. & M.S. in Computer Science

University of California San Diego, USA

Sep 2018–Apr 2023

Advisor: Taylor Berg-Kirkpatrick. M.S. awarded Jun 2020. CGPA 3.90/4.00.

B.Sc. in Computer Engineering

Sharif University of Technology, Iran

Sep 2014–Jun 2018

AWARDS & HONORS

Delta Institute Scientific Fellow	2026
Greenwall Foundation Faculty Scholars Program in Bioethics, Finalist	2026
Momental Foundation Mistletoe Research Fellowship (MRF), Finalist	2023
Rising Star in Adversarial Machine Learning (AdvML), Award Winner	2022
Rising Stars in EECS	2022
UCSD CSE Excellence in Leadership and Service, Award Winner	2022
FAccT Doctoral Consortium	2022
Qualcomm Innovation Fellowship, Finalist	2021
NCWIT (National Center for Women & IT) Collegiate Award, Winner	2020
National University Entrance Exam in Math (Ranked 249 th of 223,000)	2014
National University Entrance Exam in Foreign Languages (Ranked 57 th of 119,000)	2014
National Organization for Exceptional Talents (NODET), Admitted (~2% Acceptance Rate)	2008

GRANTS & FUNDING

<i>Research Grants & Gifts</i>	
DOE Genesis Mission — collaboration with Brookhaven National Laboratory	2026
Longview Philanthropy — AI hardware security & verification of international AI agreements	2026
Fujitsu — research grant	2026
OpenAI — Mental Health Research Grant	2026
Anthropic — AI for Science Program	2026
Foresight Institute — AI for Science & Safety	2026
<i>Compute & Platform Grants</i>	
Google TPU Research Cloud (TRC)	2026
Google Gemini compute grant	2026
Thinking Machines (Tinker) Academic Research Compute Grant	2025
Modal Academic Research Compute Grant	2025
Additional compute support: Lambda, River, Prime Intellect Academic Research Compute Grants	2025–2026

PUBLICATIONS

* denotes equal contribution. For the full list, see Google Scholar.

Preprints & Under Review

1. A. Priyanshu, S. Vijay, B. Jabarian, and **N. Mireshghallah**, “Et Tu, Brute? Economic Misalignment in Personal AI Agents,” *arXiv:2609.24927*, 2026.
2. S. Majidi, **N. Mireshghallah**, and K. Taram, “Leaky Language Models: Stealing Architecture and Inference Optimizations via Per-Token Timing,” *arXiv:2607.20723*, 2026.
3. A. Hughes, A. Goldberg, P. Jha, A. Perer, N. Aletras, and **N. Mireshghallah**, “Boundary-Targeted Membership Inference Attacks on Safety Classifiers,” *arXiv:2605.22373*, 2026.

4. K. Han, R. Zhang, K. Wei, H. Mahdavi, **N. Mireshghallah**, and A. Barati Farimani, "SMDD-Bench: Can LLMs Solve Real-World Small Molecule Drug Design Tasks?," *arXiv:2605.21740*, 2026. Poster at the GenBio Workshop, ICML 2026.
5. K. Monteiro, M. Park, A. Ioffrida, A. Sanna, H.-P. Lee, **N. Mireshghallah**, Y. Wang, and S. Das, "When Are LLM Inferences Acceptable? User Reactions and Control Preferences for Inferred Personal Information," *arXiv:2605.10013*, 2026.
6. X. Liu, **N. Mireshghallah**, J. C. Ginsburg, and T. Chakrabarty, "Alignment Whack-a-Mole: Finetuning Activates Verbatim Recall of Copyrighted Books in Large Language Models," *arXiv:2603.20957*, 2026. **Oral** at the MemFM Workshop, ICML 2026.
7. J. X. Morris, **N. Mireshghallah**, M. Ibrahim, and S. Mahloujifar, "Learning to Reason in 13 Parameters," *arXiv:2602.04118*, 2026.
8. J. Borkar, K. Chadha, **N. Mireshghallah**, Y. Zhang, I.-E. Veliche, A. Mitra, D. A. Smith, Z. Xu, and D. Garcia-Olano, "Memorization Dynamics in Knowledge Distillation for Language Models," *arXiv:2601.15394*, 2026.
9. R. Schaeffer, J. Kazdan, B. Abbasi, K. Z. Liu, B. Miranda, A. Ahmed, F. Barez, A. Puri, S. Biderman, **N. Mireshghallah**, and S. Koyejo, "Quantifying the Effect of Test Set Contamination on Generative Evaluations," *arXiv:2601.04301*, 2026. Poster at the FoGen Workshop, ICML 2026.
10. P. Mahdavinia, H. Mahdavi, **N. Mireshghallah**, and M. Mahdavi, "Harnessing Optimization Dynamics for Curvature-Informed Model Merging," *arXiv:2509.11167*, 2025.
11. J. Roh, Z. Novack, Y. Peng, **N. Mireshghallah**, T. Berg-Kirkpatrick, and A. Houmansadr, "Bob's Confetti: Phonetic Memorization Attacks in Music and Video Generation," *arXiv:2507.17937*, 2025.
12. A. Naseh and **N. Mireshghallah**, "Synthetic Data Can Mislead Evaluations: Membership Inference as Machine Text Detection," *arXiv:2501.11786*, 2025.
13. M. H. Garcia, A. Manoel, D. M. Diaz, **N. Mireshghallah**, R. Sim, and D. Dimitriadis, "FLUTE: A Scalable, Extensible Framework for High-Performance Federated Learning Simulations," *arXiv:2203.13789*, 2022.
14. **N. Mireshghallah**, M. Taram, P. Vepakomma, A. Singh, R. Raskar, and H. Esmailzadeh, "Privacy in Deep Learning: A Survey," *arXiv:2004.12254*, 2020.

Conference Papers

15. R. Zhang, M. Kaniselman, and **N. Mireshghallah**, "Reinforcement Learning Improves Traversal of Parametric Knowledge in LLMs," *EMNLP 2026 (Oral)*.
16. H. Kim*, **N. Mireshghallah***, M. Duan, R. Xin, S. S. Li, J. Jung, D. Acuna, Q. Pang, H. Xiao, G. E. Suh, S. Oh, Y. Tsvetkov, P. W. Koh, and Y. Choi, "Privasis: Synthesizing the Largest 'Public' Private Dataset from Scratch," *ICML 2026*.
17. S. Biderman, M. A. Khan, **N. Mireshghallah**, C. Arnett, F. Barez, and N. Saphra, "Position: Don't Just 'Fix it in Post': A Science of AI Must Study Learning Dynamics," *ICML 2026 (Oral)*.
18. **N. Mireshghallah**, N. Mangaokar, N. Kokhlikyan, A. Zharmagambetov, M. Zaheer, S. Mahloujifar, and K. Chaudhuri, "CIMemories: A Compositional Benchmark for Contextual Integrity of Persistent Memory in LLMs," *ICLR 2026*.
19. J. Zhou, **N. Mireshghallah**, and T. Li, "Operationalizing Data Minimization for Privacy-Preserving LLM Prompting," *ICLR 2026*.
20. T. Sorensen, B. Newman, J. Moore, C. Y. Park, J. Fisher, **N. Mireshghallah**, L. Jiang, and Y. Choi, "Spectrum Tuning: Post-Training for Distributional Coverage and In-Context Steerability," *ICLR 2026*.
21. D. Pham, P. Kairouz, **N. Mireshghallah**, E. Bagdasarian, C. M. Pham, and A. Houmansadr, "Can Large Language Models Really Recognize Your Name?," *FAccT 2026*.
22. R. Xin*, **N. Mireshghallah***, S. S. Li, M. Duan, H. Kim, Y. Choi, Y. Tsvetkov, S. Oh, and P. W. Koh, "A False Sense of Privacy: Evaluating Textual Data Sanitization Beyond Surface-Level Privacy Leakage," *SaTML 2026*.

23. A. F. Cooper, ..., **N. Mireshghallah**, ..., and K. Lee, "Machine Unlearning Doesn't Do What You Think: Lessons for Generative AI Policy and Research," *NeurIPS 2025 (Oral)*.
24. J. Hayes, I. Shumailov, C. A. Choquette-Choo, M. Jagielski, G. Kaissis, M. Nasr, M. S. M. S. Annamalai, **N. Mireshghallah**, I. Shilov, M. Meeus, Y.-A. de Montjoye, K. Lee, F. Boenisch, A. Dziedzic, and A. F. Cooper, "Exploring the Limits of Strong Membership Inference Attacks on Large Language Models," *NeurIPS 2025*.
25. T. Chen, F. Brahman, J. Liu, **N. Mireshghallah**, W. Shi, P. W. Koh, L. Zettlemoyer, and H. Hajishirzi, "ParaPO: Aligning Language Models to Reduce Verbatim Reproduction of Pre-training Data," *COLM 2025*.
26. S. Hallinan, J. Jung, M. Sclar, X. Lu, A. Ravichander, S. Ramnath, Y. Choi, S. P. Karimireddy, **N. Mireshghallah**, and X. Ren, "The Surprising Effectiveness of Membership Inference with Simple N-Gram Coverage," *COLM 2025*.
27. X. Zhou, H. Kim, F. Brahman, L. Jiang, H. Zhu, X. Lu, F. Xu, B. Y. Lin, Y. Choi, **N. Mireshghallah**, R. Le Bras, and M. Sap, "HAICOSYSTEM: An Ecosystem for Sandboxing Safety Risks in Interactive AI Agents," *COLM 2025*.
28. J. Borkar, M. Jagielski, K. Lee, **N. Mireshghallah**, D. A. Smith, and C. A. Choquette-Choo, "Privacy Ripple Effects from Adding or Removing Personal Information in Language Model Training," *Findings of ACL 2025*. Cited in US Congressional testimony on AI chatbot data privacy.
29. X. Lu, M. Sclar, S. Hallinan, **N. Mireshghallah**, J. Liu, S. Han, A. Ettinger, L. Jiang, K. R. Chandu, N. Dziri, and Y. Choi, "AI as Humanity's Salieri: Quantifying Linguistic Creativity of Language Models via Systematic Attribution of Machine Text against Web Text," *ICLR 2025 (Oral)*.
30. I. C. Ngong, J. P. Near, and **N. Mireshghallah**, "Differentially Private Learning Needs Better Model Initialization and Self-Distillation," *NAACL 2025 (Oral)*.
31. A. Ravichander, J. Fisher, T. Sorensen, X. Lu, M. Antoniak, B. Y. Lin, **N. Mireshghallah**, C. Bhagavatula, and Y. Choi, "Information-Guided Identification of Training Data Imprint in (Proprietary) Large Language Models," *NAACL 2025 (Oral; Honorable Mention candidate)*.
32. A. M. Kassem*, O. Mahmoud*, **N. Mireshghallah***, H. Kim, Y. Tsvetkov, Y. Choi, S. Saad, and S. Rana, "Alpaca against Vicuna: Using LLMs to Uncover Memorization of LLMs," *NAACL 2025*.
33. L. Jiang, K. Rao, S. Han, A. Ettinger, F. Brahman, S. Kumar, **N. Mireshghallah**, X. Lu, M. Sap, Y. Choi, and N. Dziri, "WildTeaming at Scale: From In-the-Wild Jailbreaks to (Adversarially) Safer Language Models," *NeurIPS 2024*.
34. T. Chen, A. Asai*, **N. Mireshghallah***, S. Min, J. Grimmermann, Y. Choi, H. Hajishirzi, L. Zettlemoyer, and P. W. Koh, "CopyBench: Measuring Literal and Non-Literal Reproduction of Copyright-Protected Text in Language Model Generation," *EMNLP 2024*.
35. M. Duan, A. Suri, **N. Mireshghallah**, S. Min, W. Shi, L. Zettlemoyer, Y. Tsvetkov, Y. Choi, D. Evans, and H. Hajishirzi, "Do Membership Inference Attacks Work on Large Language Models?," *COLM 2024*.
36. **N. Mireshghallah***, M. Antoniak*, Y. More*, Y. Choi, and G. Farnadi, "Trust No Bot: Discovering Personal Disclosures in Human-LLM Conversations in the Wild," *COLM 2024*.
37. T. Sorensen, J. Moore, J. Fisher, M. Gordon, **N. Mireshghallah**, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, T. Althoff, and Y. Choi, "Position: A Roadmap to Pluralistic Alignment," *ICML 2024*.
38. **N. Mireshghallah***, H. Kim*, X. Zhou, Y. Tsvetkov, M. Sap, R. Shokri, and Y. Choi, "Can LLMs Keep a Secret? Testing Privacy Implications of Language Models via Contextual Integrity Theory," *ICLR 2024 (Spotlight)*. Used by Anthropic and Meta as a pre-deployment privacy evaluation.
39. X. Tang, R. Shin, H. A. Inan, A. Manoel, **N. Mireshghallah**, Z. Lin, S. Gopi, J. Kulkarni, and R. Sim, "Privacy-Preserving In-Context Learning with Differentially Private Few-Shot Generation," *ICLR 2024*.
40. M. Zhang, T. He, T. Wang, L. Mi, **N. Mireshghallah**, B. Chen, H. Wang, and Y. Tsvetkov, "LatticeGen: Hiding Generated Text in a Lattice for Privacy-Aware Large Language Model Generation on Cloud," *Findings of NAACL 2024*.
41. **N. Mireshghallah**, J. Mattern, S. Gao, R. Shokri, and T. Berg-Kirkpatrick, "Smaller Language Models are Better Zero-shot Machine-Generated Text Detectors," *EACL 2024*.

42. **N. Mireshghallah**, Y. Su, T. Hashimoto, J. Eisner, and R. Shin, "Privacy-Preserving Domain Adaptation of Semantic Parsers," *ACL* 2023.
43. J. Mattern, **N. Mireshghallah**, Z. Jin, B. Schölkopf, M. Sachan, and T. Berg-Kirkpatrick, "Membership Inference Attacks against Language Models via Neighbourhood Comparison," *Findings of ACL* 2023.
44. **N. Mireshghallah**^{*}, N. Vogler^{*}, J. He, O. Florez, A. El-Kishky, and T. Berg-Kirkpatrick, "Simple Temporal Adaptation to Changing Label Sets: Hashtag Prediction via Dense KNN," *EMNLP* 2023.
45. J. Forristal, **N. Mireshghallah**, G. Durrett, and T. Berg-Kirkpatrick, "A Block Metropolis-Hastings Sampler for Controllable Energy-Based Text Generation," *CoNLL* 2023.
46. **N. Mireshghallah**, A. Backurs, H. A. Inan, L. Wutschitz, and J. Kulkarni, "Differentially Private Model Compression," *NeurIPS* 2022.
47. **N. Mireshghallah**, A. Uniyal, T. Wang, D. Evans, and T. Berg-Kirkpatrick, "An Empirical Analysis of Memorization in Fine-tuned Autoregressive Language Models," *EMNLP* 2022 (**Oral**).
48. **N. Mireshghallah**, K. Goyal, A. Uniyal, T. Berg-Kirkpatrick, and R. Shokri, "Quantifying Privacy Risks of Masked Language Models Using Membership Inference Attacks," *EMNLP* 2022.
49. H. Brown, K. Lee, **N. Mireshghallah**, R. Shokri, and F. Tramèr, "What Does it Mean for a Language Model to Preserve Privacy?," *FAccT* 2022.
50. **N. Mireshghallah**, V. Shrivastava, M. Shokouhi, T. Berg-Kirkpatrick, R. Sim, and D. Dimitriadis, "UserIdentifier: Implicit User Representations for Simple and Effective Personalized Sentiment Analysis," *NAACL* 2022.
51. **N. Mireshghallah**, K. Goyal, and T. Berg-Kirkpatrick, "Mix and Match: Learning-free Controllable Text Generation using Energy Language Models," *ACL* 2022.
52. **N. Mireshghallah** and T. Berg-Kirkpatrick, "Style Pooling: Automatic Text Style Obfuscation for Improved Classification Fairness," *EMNLP* 2021 (**Oral**).
53. **N. Mireshghallah**, H. A. Inan, M. Hasegawa, V. Rühle, T. Berg-Kirkpatrick, and R. Sim, "Privacy Regularization: Joint Privacy-Utility Optimization in Language Models," *NAACL* 2021.
54. **N. Mireshghallah**, M. Taram, A. Jalali, A. T. Elthakeb, D. Tullsen, and H. Esmailzadeh, "Not All Features Are Equal: Discovering Essential Features for Preserving Prediction Privacy," *WWW* 2021.
55. T. Koker, **N. Mireshghallah**, T. Titcombe, and G. Kaissis, "U-Noise: Learnable Noise Masks for Interpretable Image Segmentation," *ICIP* 2021.
56. A. T. Elthakeb, P. Pilligundla, **N. Mireshghallah**, A. Cloninger, and H. Esmailzadeh, "Divide and Conquer: Leveraging Intermediate Feature Representations for Quantized Training of Neural Networks," *ICML* 2020.
57. **N. Mireshghallah**, M. Taram, P. Ramrakhiani, A. Jalali, D. Tullsen, and H. Esmailzadeh, "Shredder: Learning Noise Distributions to Protect Inference Privacy," *ASPLOS* 2020.

Journal Papers

58. A. T. Elthakeb, P. Pilligundla, **N. Mireshghallah**, A. Yazdanbakhsh, and H. Esmailzadeh, "ReLeQ: A Reinforcement Learning Approach for Automatic Deep Quantization of Neural Networks," *IEEE Micro*, 2020.
59. **N. Mireshghallah**, M. Bakhshalipour, M. Sadrosadati, and H. Sarbazi-Azad, "Energy-Efficient Permanent Fault Tolerance in Hard Real-Time Systems," *IEEE Transactions on Computers*, 2019.

Workshop & Tutorial Papers

60. R. Zhang and **N. Mireshghallah**, "Reinforcement Learning on Benign Facts Amplifies Leakage of Memorized Private Data," *TAE Workshop at NeurIPS* 2026. arXiv:2608.21727.
61. E. Farhana, M. Zainab, Q. Wang, **N. Mireshghallah**, R. van der Meulen, and M. J. van Duijn, "Theory of Mind and Application in Educational Context," *BEA Workshop at ACL* 2026 (tutorial).

62. H. Mahdavi, P. Mahdavinia, S. Malek, P. Mohammadipour, A. Hashemi, M. Daliri, A. Farhadi, A. Khasahmadi, **N. Mireshghallah**, and V. G. Honavar, "RefGrader: Automated Grading of Mathematical Competition Proofs using Agentic Workflows," *NeurIPS 2025 Workshop on MATH-AI*.
63. Y. Bae, M. Kim, J. Lee, S. Kim, J. Kim, Y. Choi, and **N. Mireshghallah**, "Privacy-Preserving LLM Interaction with Socratic Chain-of-Thought Reasoning and Homomorphically Encrypted Vector Databases," *Workshop on Privacy-Preserving Machine Learning (PPML) at CRYPTO 2025*.
64. N. G. Brigham, C. Gao, T. Kohno, F. Roesner, and **N. Mireshghallah**, "Breaking News: Case Studies of Generative AI's Use in Journalism," *SoLaR Workshop at NeurIPS 2024*.
65. K. Lee, A. F. Cooper, C. A. Choquette-Choo, K. Liu, M. Jagielski, **N. Mireshghallah**, et al., "Machine Unlearning Doesn't Do What You Think," *GenLaw Workshop at ICML 2024 (Extended Abstract)*.
66. R. Naidu, A. Priyanshu, A. Kumar, S. Kotti, H. Wang, and **N. Mireshghallah**, "When Differential Privacy Meets Interpretability: A Case Study," *Responsible Computer Vision Workshop at CVPR 2021*.
67. P. Basu, T. Singha Roy, R. Naidu, Z. Müftüoglu, S. Singh, and **N. Mireshghallah**, "Benchmarking Differential Privacy and Federated Learning for BERT Models," *Machine Learning for Data Workshop at ICML 2021*.
68. A. Uniyal, R. Naidu, S. Kotti, S. Singh, P. J. Kenfack, **N. Mireshghallah**, and A. Trask, "DP-SGD vs. PATE: Which Has Less Disparate Impact on Model Accuracy?," *Machine Learning for Data Workshop at ICML 2021*.
69. T. Farrand, **N. Mireshghallah**, S. Singh, and A. Trask, "Neither Private Nor Fair: Impact of Data Imbalance on Utility and Fairness in Differential Privacy," *PPMLP Workshop at CCS 2020*.

Patents

70. J. M. Eisner, E. C. Shin, **N. Mireshghallah**, T. B. Hashimoto, and Y. Su, "Privacy-Preserving Generation of Synthesized Training Data," US Patent Application 18/321,460 (2024).
71. **N. Mireshghallah** and H. Esmailzadeh, "Methods of Providing Data Privacy for Neural Network-Based Inference," US Patents 11,288,379; 11,487,884; and 12,210,971.
72. **N. Mireshghallah**, H. Esmailzadeh, and M. Taram, "Methods and Systems for Learning Noise Distributions for Neural Networks," US Patent 12,008,470 (granted Jun. 2024).

Policy & Technical Write-ups

73. **N. Mireshghallah** and T. Li, "Position: Privacy Is Not Just Memorization!," *arXiv:2510.01645*, 2025. Featured in Help Net Security.
74. A. F. Cooper, K. Lee, J. Grimmelmann, D. Ippolito, C. Callison-Burch, C. A. Choquette-Choo, **N. Mireshghallah**, et al., "Report of the 1st Workshop on Generative AI and Law," *Yale Law & Economics Research Paper, SSRN-4634513*, 2023.

INVITED TALKS

Keynotes & Recent Invited Talks

Carnegie Mellon University, School of Computer Science

New Faculty Lightning Talk

Learning the Tails!

Aug. 2026

ICML 2026 (Seoul)

Invited Debater, Workshop on AI as a Tool for Mathematics, CS & ML

Debate with Ludwig Schmidt: will society keep paying humans to do research?

Jul. 2026

ICML 2026 (Seoul)

Panelist, WiML Symposium ("Career Transitions in ML") and KAIST@ICML ("Career Advice for Junior Researchers")
Jul. 2026

Global AI Frontiers Symposium (Seoul)

Panelist
Jul. 2026

BEA Workshop at ACL 2026

Tutorial Co-instructor
Theory of Mind and Application in Educational Context.
Jul. 2026

Google

Invited Talk
Jun. 2026

University of Cagliari (PRA Lab), Machine Learning Security Seminar Series

Invited Talk
What Should an Agent Remember, Reveal, and Redact? The New Data Sinks and Leaks of Agentic AI.
Jun. 2026

Cornell University

AI Seminar
2026 Is the New 2016, but Make It Privacy: On Federated Memory, Contextual Privacy, and Personalized Agents.
May 2026

MIT CSAIL

NLP Seminar
2026 Is the New 2016, but Make It Privacy: On Federated Memory, Contextual Privacy, and Personalized Agents.
Apr. 2026

Simons Institute, UC Berkeley

Keynote, Workshop on Trust in Decentralized Systems
2026 Is the New 2016, but Make It Privacy: On Federated Memory, Contextual Privacy, and Personalized Agents.
Mar. 2026

FAR AI San Diego Alignment Workshop at NeurIPS

Invited Talk
What Does It Mean for Agentic AI to Preserve Privacy?
Dec. 2025

Conference on Applied Machine Learning in Information Security (CAMLIS)

Keynote
What Does It Mean for Agentic AI to Preserve Privacy? Mapping the New Data Sinks and Leaks.
Oct. 2025

Cornell Tech

Digital Life Seminar
Contextual Privacy in LLMs: Benchmarking and Mitigating Inference-Time Risks.
Oct. 2025

Meta AI / FAIR Alignment Group

Invited Talk
What You Should Really Worry About When It Comes to Generative AI and Privacy.
Oct. 2025

First Workshop on LLM Security (LLMSec) at ACL 2025 <i>Keynote</i> What Does It Mean for Agentic AI to Preserve Privacy?	<i>Aug. 2025</i>
First Workshop on LLM Memorization (L2M2) at ACL 2025 <i>Keynote</i> Emergent Misalignment Through the Lens of Non-verbatim Memorization.	<i>Aug. 2025</i>
Workshop on Collaborative and Federated Agentic Workflows (CFAgentic) at ICML 2025 <i>Invited Talk</i>	<i>Jul. 2025</i>
Fifth Workshop on Trustworthy NLP (TrustNLP) at NAACL 2025 <i>Invited Talk</i> A False Sense of Privacy: Semantic Leakage and Non-literal Copying in LLMs.	<i>May 2025</i>
ETH Zurich <i>Guest Lecture</i> Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.	<i>May 2025</i>
UC Berkeley School of Information <i>Invited Talk</i> Privacy, Copyright, and Data Integrity: The Cascading Implications of Generative AI.	<i>Feb. 2025</i>
Stanford University <i>NLP Seminar</i> Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.	<i>Jan. 2025</i>
Academic Job Talks (Jan.–Mar. 2025) <i>Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.</i> Johns Hopkins University, UC Berkeley, Carnegie Mellon University, UW–Madison, UPenn, Georgia Tech, UCLA, University of Maryland, University of Michigan, NYU, EPFL, ETH Zurich, UT Austin, UVA, UNC Chapel Hill.	
2024	
NeurIPS 2024 Red Teaming GenAI Workshop <i>Keynote</i> A False Sense of Privacy: Semantic Leakage and Non-literal Copying in LLMs.	<i>Dec. 2024</i>
NeurIPS 2024 PrivacyML Tutorial <i>Panelist</i>	<i>Dec. 2024</i>
Johns Hopkins University <i>CS Department Seminar</i> Privacy, Copyright and Data Integrity: The Cascading Implications of Generative AI.	<i>Dec. 2024</i>
Future of Privacy Forum <i>Panelist, Technologist Roundtable for Policymakers</i> Key Issues in Privacy and AI.	<i>Nov. 2024</i>
National Academies (NASEM)	

University seminars & guest lectures (2024): UMass Amherst NLP Seminar; Northeastern Khoury Security Seminar; UC Santa Barbara & UCLA NLP Seminars; Georgia Tech School of Interactive Computing; University of Utah; UW Allen School Colloquium; UT Austin; CMU (LTI 11-830 & invited talk).

Industry & research-org talks (2024): Google Brain; Meta AI Research; LinkedIn Research; SRI International C3E Workshop; Simons Collaboration TOC4Fairness Seminar; Generative AI & Law Workshop (Washington, DC); ML Collective DLCT.

Earlier Talks (selected)
CISPA Helmholtz Center for Information Security (2023); Max Planck Institute for Software Systems (2023); NDSS EthiCS Workshop — Keynote (2023); Google Federated Learning Seminar (2023); Mila / McGill University (2023); EACL 2023 Tutorial Co-instructor; UW eScience Institute (2023); Cohere for AI (2023).
KDD AdvML Workshop (2022); UT Austin (2022); Microsoft Research Cambridge (2022); Johns Hopkins University Guest Lecture (2022); PriSec ML Interest Group (2022).
Alan Turing Institute (2021); National University of Singapore (2021); Big Science for LLMs Panel (2021); Split Learning Workshop (2021); University of Maine Guest Lecture (2021).
UMass Amherst ML and Friends Lunch (2020); OpenMined Privacy Conference (2020); Microsoft Research AI Breakthroughs Workshop (2020); NCWIT Collegiate Award Winner Talk (2020).

STUDENT MENTORING & ADVISING

Current — Looni Lab, Carnegie Mellon University
Bill Watson, Ph.D. student (co-advised with Sauvik Das)
Xiaoyu (Nicholas) Wu, Ph.D. student (co-advised with Steven Wu)
Renfei Zhang, Ph.D. student
Kevin Han, Ph.D. student (co-advised with Amir Barati Farimani)
Bhuvan Chandra Koduru, Master's student
Visiting researchers & student collaborators: Prince Jha, Anthony Hughes

Past Mentees (2020–2025)
Affiliations as of the mentorship period; arrows show subsequent placement.
Skyler Hallinan, BS/MS student at University of Washington → PhD student at USC
Tong Chen, PhD student at University of Washington
Sara Kodeiri, MS student at University of Waterloo
Sabrina Mokhtari, MS student at University of Waterloo
Rui Xin, PhD student at University of Washington
Yash More, MS student at McGill University
Aly Kassem, MS student at University of Windsor → Research Engineer at Dynamo AI
Michael Duan, BS student at University of Washington → Software Engineer at Uber
Grace Brigham, BS student at University of Washington
Chongjiu Gao, BS/MS student at University of Washington
Supriti Vijay, BS at Manipal Institute → MS student at Carnegie Mellon University
Justus Mattern, BS student at RWTH Aachen → Y Combinator
Ivoline Ngong, PhD student at University of Vermont
Jarad Forristal, BS at University of Texas Austin → PhD student at UC San Diego
Archit Uniyal, BE at UIET Panjab University → PhD student at University of Virginia
Aman Priyanshu, BS at Manipal Institute → MS student at Carnegie Mellon University
Rakshit Naidu, BS at Manipal Institute → PhD student at Georgia Tech

Priyam Basu, BS at Manipal Institute → MS student at University of Washington
Zümrüt Müftüoglu, PhD student at Yıldız Teknik Üniversitesi
Teddy Koker, Research Engineer at Lightning AI → PhD student at MIT EECS
Tom Farrand, Deep Learning Engineer at IBM
Jaydeep Borkar, BE at Pune University → PhD student at Northeastern University

TEACHING

CSE Department, UC San Diego (Teaching Assistant)

CSE 151A (Undergraduate ML); CSE 251A (Graduate ML); CSE 276C (Graduate Mathematics for Robotics); CSE 141 (Undergraduate Computer Architecture); CSE 240D (Graduate Accelerator Design for Deep Learning) 2019–2021 **CE**

Department, Sharif University of Technology (Teaching Assistant)

Digital Electronics; Computer Architecture; Signals and Systems; Probability and Statistics; Numerical Methods 2016–2018

ORGANIZED EVENTS & WORKSHOPS

Co-organizer, 2nd Workshop on Memorization and Trustworthy Foundation Models (MemFM) at ICML	2026
Co-organizer, Workshop on Foundations of Deep Generative Models (FoGen) at ICML	2026
Advisory Committee, 2nd Workshop on Technical AI Governance Research (TAIGR) at ICML	2026
Co-organizer, Memorization and Trustworthy Foundation Models (MemFM) Workshop at ICML	2025
Privacy Session Chair, SAGAI Workshop at IEEE S&P	2025
Co-organizer, Generative AI and Law (GenLaw) Workshop at ICML	2024
Co-organizer, Privacy Regulation and Protection in Machine Learning Workshop at ICLR	2024
Co-organizer, Private NLP Workshop at ACL	2024
Co-organizer, Privacy-Preserving AI (PPAI) Workshop at AAAI	2024
Co-organizer, Generative AI and Law (GenLaw) Workshop at ICML	2023
Co-organizer, Widening NLP (WiNLP) Workshop at EMNLP	2022, 2023
Co-organizer, Private NLP Tutorial at EACL	2023
Co-organizer, Ethics in NLP Birds of a Feather Session at EMNLP	2022
Privacy & Fairness Roundtable Lead, AFCEP Workshop at NeurIPS	2022
Co-organizer, Broadening Collaborations in ML Workshop at NeurIPS	2022
Co-organizer, Private NLP Workshop at NAACL	2022
Co-organizer, Federated Learning for NLP Workshop at ACL	2022
Co-organizer, Widening NLP (WiNLP) Workshop at EMNLP	2021
Co-organizer, Privacy-Preserving Machine Learning (PPML) Workshop at MICCAI	2021
Co-organizer, Distributed Private Machine Learning (DPML) Workshop at ICLR	2021
Co-organizer, SoCal Joint ML and NLP Symposium	2021
Break-out Co-leader, Women in Machine Learning (WiML) Un-workshop at ICML	2020, 2021

PROFESSIONAL SERVICE

Area Chair, NeurIPS	2026
Area Chair, COLM	2025, 2026
Senior Program Committee, AAAI Alignment Track	2027
Publicity Chair, ACL	2027
Area Chair, NAACL	2025
Area Chair, EMNLP	2024
PC Member, ACM CCS	2024, 2025
D&I Co-chair, NAACL	2022, 2025
Co-chair, Widening NLP (WiNLP)	2022–2024
Reviewer, ICLR (Outstanding Reviewer Award, 2021)	2021–2025
Reviewer, NeurIPS; ICML	2020–2024

Reviewer, AISTATS	2025
Reviewer, ICLR & ICML Workshop Proposals	2025
Reviewer, COLM; EACL; CHI	2024
Reviewer, AAAI; FAccT	2023, 2024
Reviewer, TACL	2020–Present
Reviewer, TMLR	2022–Present
Reviewer, IEEE S&P Magazine	2021–Present
Reviewer, NeurIPS Workshop Proposals; INLG	2023
Ethics PC Member, EMNLP; Reviewer, NAACL SRW; PC Member, AFCEP & TSRML Workshops at NeurIPS	2022
Shadow PC Member, IEEE S&P; Artifact Evaluation PC, USENIX Security; Reviewer, CCS Posters	2021
Artifact Evaluation PC, ASPLOS; PC Member, LatinX in AI & WHI Workshops at ICML, MLArchSys at ISCA; Security & Privacy Committee Member and Session Chair, Grace Hopper Celebration	2020
Reviewer, ACM TACO and IEEE TC Journals	2020–Present

DIVERSITY & INCLUSION

Mentor, Women in Machine Learning (WiML) Workshop at NeurIPS	2020, 2024, 2025
Panelist, CMU SCS Panel: Navigating the Academic Job Market	2025
Mentor, ACL Mentorship: How to Broadcast Your Research to a Wider Audience?	2025
Mentor, Graduate Women in Computing (GradWIC) at UCSD	2020–2023
Mentor, UC San Diego Women Organization for Research Mentoring (WORM) in STEM	2019–2023
Course Instructor, OpenMined Privacy Course	2020
Mentor, USENIX Security Undergraduate Mentorship Program	2020
Mentor, ICLR	2021
Volunteer & Invited Speaker, WiML Workshop and WiMLDS Meetup at NeurIPS	2019
Mentor, UCSD CSE Early Research Scholars Program (CSE-ERSP)	2018

FEATURED PRESS & MEDIA

BBC Persian, <i>60 Minutes</i> news program — live segment on AI risks and society	Sep. 2026
MIT Technology Review — “We still don’t know how people are really using AI” (coverage of the AI Observatory)	Aug. 2026
CMU News — “Carnegie Mellon Researchers Lead Three DOE Genesis Mission Awards to Advance the Future of AI-Enabled Scientific Discovery”	Jul. 2026
The Information Bottleneck podcast — on the future of generative AI	Jan. 2026
Science News Explores — “5 things to remember when talking to a chatbot”	Nov. 2025
Help Net Security — “Most AI privacy research looks the wrong way”	Oct. 2025
Washington Post — on AI hype, evaluation metrics, and how people judge AI capabilities	Aug. 2025
Jay Shah Podcast — “Differential Privacy, Creativity & Future of AI Research in the LLM Era”	Feb. 2025
Science — “AI writing is improving, but it still can’t match human creativity”	Dec. 2024
The Thesis Review podcast (with Sean Welleck) — on auditing and mitigating safety risks in LLMs	2024
Washington Post — on Google AI overview errors, chatbot usage patterns, and opting out of AI training	2024
WIRED — “How to Stop Your Data From Being Used to Train AI”	2024
UW News — on advances in math and reasoning in new ChatGPT versions	2024